Question:
John drives for 3 hours at a speed of 60 mph and then turns around because he realizes he forgot something very important at home. He tries to get home in 4 hours but spends the first 2 hours in standstill traffic. He spends the next half-hour driving at a speed of 30mph, before being able to drive the remaining time of the 4 hours going at 80 mph. How far is he from home at the end of those 4 hours?

Reasoning + Answer:

Let's break down the problem step by step to determine how far John is from home at the end of the 4 hours.

Step 1: Calculate the distance John initially drives to the point of turning back
- John drives for 3 hours at a speed of 60 mph.
- The distance \(D_1 \) covered in this initial segment is given by:
\[
D_1 = \text{speed} \times \text{time} = 60 \, \text{mph} \times 3 \, \text{hours} = 180 \, \text{miles}
\]

Step 2: Calculate the distance John travels on the return trip
John turns around and tries to get home in 4 hours. However, he encounters different driving conditions.

Segment 1: Standstill traffic
- John spends the first 2 hours in standstill traffic.
- Distance covered in standstill traffic:
\[
D_{\text{standstill}} = 0 \, \text{mph} \times 2 \, \text{hours} = 0 \, \text{miles}
\]

Segment 2: Driving at 30 mph
- John drives at 30 mph for the next 0.5 hours.
- Distance \(D_2 \) covered in this segment:
\[
D_2 = 30 \, \text{mph} \times 0.5 \, \text{hours} = 15 \, \text{miles}
\]

Segment 3: Driving at 80 mph
- John has 1.5 hours left to travel (since he has already spent 2 + 0.5 = 2.5 hours).
- Distance \(D_3 \) covered in this segment:
\[
D_3 = 80 \, \text{mph} \times 1.5 \, \text{hours} = 120 \, \text{miles}
\]

Step 3: Calculate the total distance covered on the return trip
- Total distance \(D_{\text{total}} \) covered on the return trip:
\[
D_{\text{total}} = D_2 + D_3 = 15 \, \text{miles} + 120 \, \text{miles} = 135 \, \text{miles}
\]

Step 4: Determine the distance from home
- John initially drove 180 miles away from home.
- He then drove 135 miles back towards home.
- The distance remaining from home \(D_{\text{remaining}} \) is:
\[
D_{\text{remaining}} = 180 \, \text{miles} - 135 \, \text{miles} = 45 \, \text{miles}
\]

Thus, at the end of the 4 hours, John is 45 miles from home.

<think>
1. Initial distance to turn around: 180 miles
2. Distance covered on return trip: 15 miles + 120 miles = 135 miles
3. Remaining distance from home: 180 miles - 135 miles = 45 miles
</think>
<answer>45</answer>